T Empirical Analysis on Additional Datasets

Empirical analysis of intermediate representations with similarity measures can help uncover the intricate workings of various neural network architectures. When two networks that are structurally identical but trained from different initial conditions are compared, we expect that, for each layer's intermediate representation, the most similar representation in the other model should be the corresponding layer that matches in structure. We can also perform cross-layer analysis and cross-downstream task analysis of the intermediate representations of a neural network to help uncover learning patterns of a network. We use 4 different GNN architectures; GCN (Kipf & Welling, 2017), GraphSAGE (Hamilton et al., 2017), ClusterGCN (Chiang et al., 2019) and Graph Attention Network (GAT) (Veličković et al., 2018). We evaluate with Sanity Tests, Cross Architecture Tests, Downstream Task tests and Convergence Tests. We utilize the Amazon Computers Dataset (Shchur et al., 2019), Cora (McCallum et al., 2000), Citeseer (Giles et al., 1998), Pubmed (Sen et al., 2008) and Flickr (Zeng et al., 2020) Datasets for our experiments.

T.1 Initialization Sanity Tests

The sanity test compares the intermediate representations obtained from each layer of the Graph Neural Network against all the other layers of another GNN with the exact same architecture but trained with a different initialization. we expect that, for each layer's intermediate representation, the most similar intermediate representation in the other model should be the corresponding layer that matches in structure. We only demonstrate the performance of NSA for both Node Classification and Link Prediction.

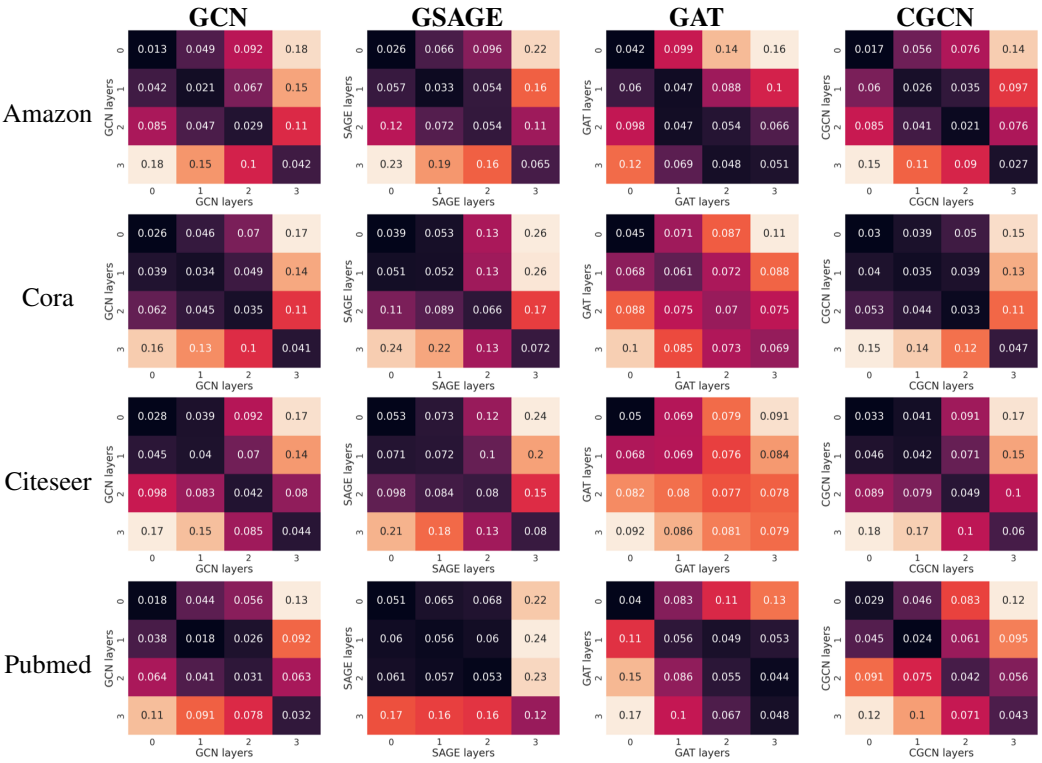


Figure 13: Sanity Tests for Link Prediction on 3 datasets. The heatmaps show layer-wise dissimilarity values for two different initialization of the same dataset on four different GNN architectures.

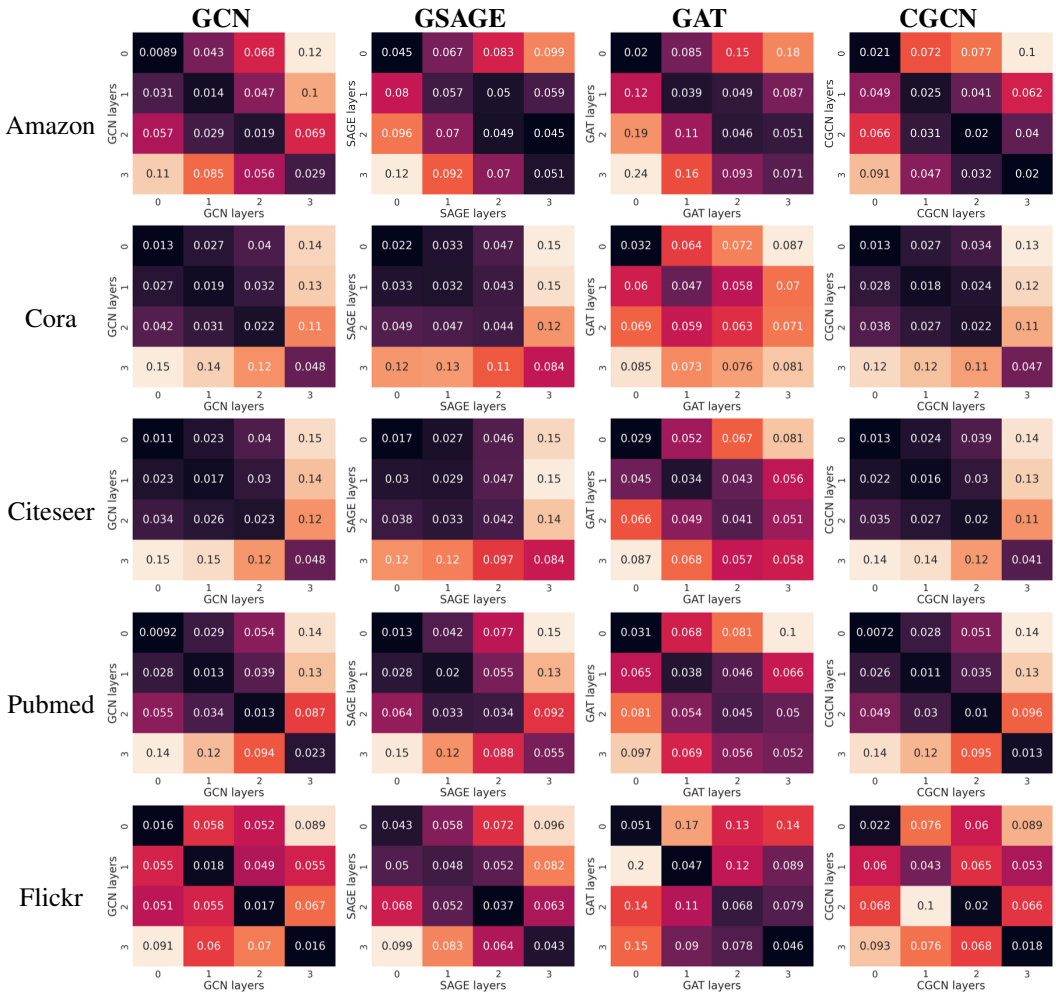
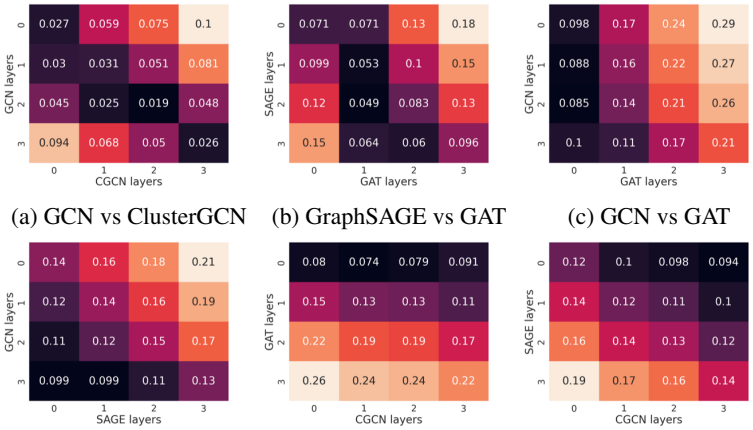


Figure 14: Sanity Tests for Node Classification on all 4 datasets. The heatmaps show layer-wise dissimilarity values for two different initialization of the same dataset on four GNN architectures.

T.2 Cross Architecture Tests on Node Classification

In this section we evaluate the similarity in representations across architectures for the Amazon Computers dataset and the three Planetoid datasets; Cora, Citeseer and Pubmed on Node Classification.

Normalized Space Alignment

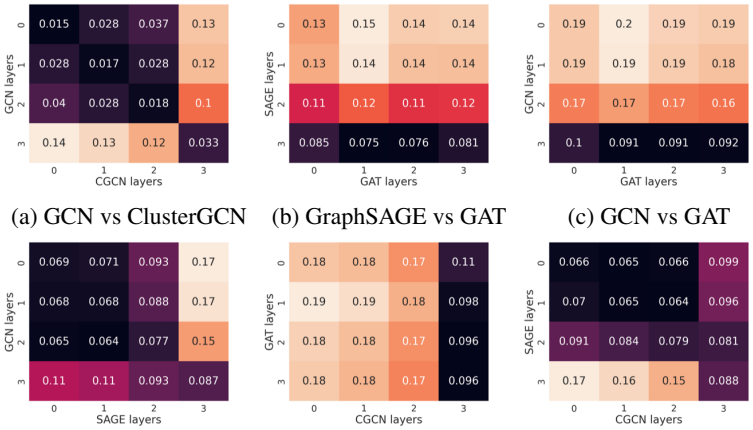


(a) GCN vs ClusterGCN (b) GraphSAGE vs GAT (c) GCN vs GAT

(d) GCN vs GraphSAGE (e) GAT vs ClusterGCN (f) GSAGE vs CGCN

Figure 15: Cross Architecture Tests using NSA for Node Classification on the Amazon Dataset

Normalized Space Alignment



(a) GCN vs ClusterGCN (b) GraphSAGE vs GAT (c) GCN vs GAT

(d) GCN vs GraphSAGE (e) GAT vs ClusterGCN (f) GSAGE vs CGCN

Figure 16: Cross Architecture Tests using NSA for Node Classification on the Cora Dataset